# ITR: Learning and Measuring Perceptual Similarity

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## Abstract

Image retrieval has been an active research area for many years, but two fundamental problems remain largely unsolved: 1) how best to learn users' query concepts (search profiles), and 2) how to measure perceptual similarity with significant accuracy. The first problem concerns the completeness of formulating a search profile, e.g., how to formulate a query such as "animals" or "enemy vessels." The second problem concerns search accuracy, i.e., given a search profile, how to find all images that match that profile.

To tackle these two fundamental problems and to ensure that our solutions are scalable, this project's specific targets are as follows:

1. To develop novel *active learning algorithms* that learn a query profile quickly despite time and sample constraints. (E.g., Constructing an enemy-vessel profile quickly with meager data.)

2. To design *semi-automatic image annotation* and *annotation refinement* methods for assigning semantical labels to image s to support multimodality query-concept learning and retrieval.

3. To devise *perceptual distance functions* for improving visual search accuracy. Once a search profile such as "enemy vessels" is learned, we would not want to miss any matching object in the surveillance database.

4. To conduct *validation* on developed learning algorithms with experimental data provided by colleagues at various institutions (including NAVY if such an arrangement is feasible).

This project could have substantial impact upon image retrieval. One early evidence of such impact is that since May 2001, the PI has been invited to present preliminary results for the CIA, and at NSF sponsored workshops at INRIA and Harvard University; ACM and IEEE sponsored conferences; IBM Almaden and T.J. Watson Research Centers; Stanford University, UC Berkeley, UCLA, and other forums.